Platform Architecture Guide

Sample document for the MCP PDF Content Finder project.

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System Overview

The Sample Shop platform uses a layered architecture. Client applications call an API gateway, which routes requests to application services. Services communicate with PostgreSQL for transactional data and Redis for short-lived cache entries.

Authentication

Users authenticate through an OAuth 2.0 compatible identity provider. The API gateway validates access tokens before forwarding requests. Services receive the authenticated user identifier and authorization claims.

Service Layer

The checkout service coordinates pricing, inventory, and order creation. The pricing service calculates totals and discounts. The inventory service verifies stock before an order is accepted.

Database

PostgreSQL stores users, products, orders, and payment references. Database migrations are applied during deployment. Application services use connection pooling and parameterized queries.

Caching and Messaging

Redis caches frequently requested product data. Background events can be placed on a message queue so non-critical work does not delay API responses.

Observability

Services emit structured logs, metrics, and request correlation identifiers. Production alerts focus on error rates, latency, database availability, and failed authentication attempts.